**25900063** **A**

**Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

## SECOND SEMESTER

## Multi-Disciplinary Courses (MDC) - MG2MDCECT101 - PYTHON FOR ELECTRONICS

(2024 ADMISSION ONWARDS)

**Duration: 1 Hours** **Maximum Marks: 35**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Multiple Choice Questions

Answer any **35** questions

Each question carries **1** marks

1. Who developed Python programming language? [K] / [CO1]

   a). Dennis Ritchie  b). Guido van Rossum  c). James Gosling  d). Bjarne Stroustrup

2. What is the output of True and True? [U] / [CO1]

   a). TRUE()  b). FALSE()  c). None  d). Error

3. Which of the following is a valid variable name in Python? [K] / [CO1]

   a). my-var  b). my_var  c). my var  d). my.var

4. What is the result of the following code? `type(100)` [K] / [CO1]

   a). int  b). flaot

   c). list  d). string

5. What will be the output of s[1:4:-1] if s = "Python"? [K] / [CO1]

a). yt b). ht c). “’ d). hty

6. How many microSD card slots does Raspberry Pi 4 have? [U] / [CO1]

a). 2 b). 1 c). 0 d). 3

7. What is the output of 10 / 2 + 3? [U] / [CO1]

a). 8 b). 5 c). 6 d). 2

8. What type of power supply is recommended for Raspberry Pi 4? [K] / [CO1]

a). 5V 2A b). 12V 1A

c). 5V 3A USB-C d). 9V 2A

9. What is the output of None < None? [K] / [CO1]

a). TRUE() b). FALSE() c). None d). Error

10. What is the output of `print("Hello", end=" ")` and `print("World!")`? [U] / [CO1]

a). Hello World! b). HelloWorld!

c). HelloWorld! d). Error

11. What is the value of x after x = 10; x /= 2; x **= 2? [U] / [CO1]

a). 25 b). 10 c). 5 d). Error

12. What is the output of 15 ^ 7? [K] / [CO1]

a). 8 b). 15 c). 7 d). 0

13. What is the output of False or not (True and False)? [K] / [CO1]

a). TRUE() b). FALSE() c). None d). Error

14. What is the output of 5 > 3? [K] / [CO1]

a). TRUE() b). FALSE() c). None d). Error

15. What is the output of 10 % (2 + 3) ** 2? [K] / [CO1]

a). 10 b). 0 c). 1 d). 2

16. What is the result of list(range(0, -5, -1))? [K] / [CO1]

a). [0, -1, -2, -3, -4] b). [0, 1, 2, 3, 4] c). [-1, -2, -3, -4, -5] d). [0, -1, -2, -3, -4, -5]

17. Which of the following statements is true about lists in Python? [K] / [CO1]

a). Lists are immutable

b). Lists are ordered and mutable

c). Lists store only integers

d). Lists do not allow duplicates

18. What is the value of `x` after `x = 8; x >>= 2`? [U] / [CO1]

a). 2 b). 4 c). 8 d). 16

19. Which Raspberry Pi model is designed as a compact, ultra-low-cost variant? [U] / [CO1]

a). Raspberry Pi 3 b). Raspberry Pi Zero

c). Raspberry Pi 4 d). Raspberry Pi Compute Module

20. What is the index of the first character in a string? [U] / [CO1]

a). 0 b). 1 c). -1 d). Depends on string

21. Which of the following is an alternative OS for Raspberry Pi? [K] / [CO1]

a). Windows 10 b). Android c). MacOS d). Linux

22. Can a tuple contain duplicate values? [K] / [CO1]

a). Yes b). No

c). Sometimes d). Depends on elements

23. What is the primary purpose of BeagleBoard? [U] / [CO1]

a). Gaming b). Embedded system development c). Office applications d). Data mining

24. What is the output of the following code? x = 0 while x < 3: print(x) x += 1 Else: print("Else block") [U] / [CO2]

a). 0, 1, 2, Else block b). 0, 1, 2

c). Else block d). 0, 1, 2, 3, Else block

25. What is the output of "{} is {} years old".format("Alice", 25)? [U] / [CO2]

a). Alice is 25 years old b). {} is {} years old

c). Alice 25 d). Error

26. How many times will the loop run? x = 1 while x < 10: X *= 2 [U] / [CO2]

a). 4 b). 5 c). 3 d). 6

27. What will be the output of if 10 < 5: print("A") elif 5 != 5: print("B") else: print("C")? [U] / [CO2]

a). A b). B c). C d). Error

28. What is the output of the following code? for i in range(1, 5): print(i * 2) else: print("Else block") [K] / [CO2]

a). 1, 2, 3, 4, Else block b). 2, 4, 6, 8

c). Else block d). 2, 4, 6, 8, Else block

29. What is the output of the following code? for i in range(1, 4): print(i * 3) [U] / [CO2]

a). 3, 6, 9 b). 1, 2, 3

c). 3, 6 d). 1, 2, 3, 4

30. What is the correct syntax for an if-elif-else statement? [U] / [CO2]

a). if-elif-else b). if-else-elif c). elif-if-else d). else-elif-if

31. What does 'hello world'.title() return? [U] / [CO2]

a). 'hello World' b). 'Hello world'

c). 'Hello World' d). 'HELLO WORLD'

32. Which GPIO pin is commonly used as the default PWM output? [A] / [CO3]

a). GPIO 18 b). GPIO 12

c). GPIO 7 d). GPIO 22

33. Which function is used to configure a GPIO pin as an input? [A] / [CO3]

a). GPIO.setmode(GPIO.IN) b). GPIO.input_setup(pin)

c). GPIO.setup(pin, GPIO.IN) d). GPIO.config_input(pin)

34. What function is used to configure a GPIO pin as an input? [An] / [CO3]

a). GPIO.setup(pin, GPIO.IN) b). GPIO.setmode(GPIO.IN)

c). GPIO.input(pin) d). GPIO.init(pin, GPIO.IN)

35. What is the main loop function in Tkinter? [A] / [CO3]

a). mainloop() b). start() c). run() d). execute()

36. What is the primary purpose of the Label widget in Tkinter? [A] / [CO3]

a). To create buttons b). To display text or images

c). To handle user input d). To manage layouts

37. Which widget is required to display a basic window? [A] / [CO3]

a). Tk() b). Label c). Button d). Canvas

38. Which function is used to write a HIGH signal to a GPIO pin? [A] / [CO3]

a). GPIO.output(pin, GPIO.HIGH) b). GPIO.set(pin, HIGH)

c). GPIO.write(pin, HIGH) d). GPIO.on(pin)

39. Which argument in pack() aligns widgets left or right? [C] / [CO3]

a). side b). align c). anchor d). justify

40. Which variable type is used to store the value of a Checkbutton? [C] / [CO3]

a). IntVar b). StringVar c). BooleanVar d). FloatVar

**[1x35 = 35]**

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Answer any **35** questions

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   a). if-elif-else b). if-else-elif c). elif-if-else d). else-elif-if

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